Preseptal and Orbital Cellulitis

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Lecture Acknowledgement and Thanks:

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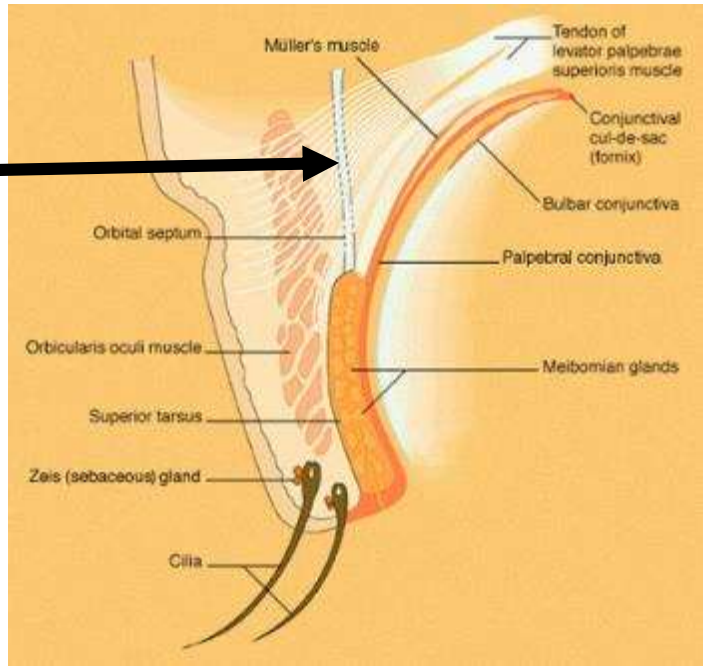
Clinical presentation	Preseptal	Orbital
Eyelid edema		
Chemosis (swelling of conj)		
Pupils/APD		
Extraocular movements +/- diplopia		
Visual acuity		
Proptosis		
Color vision		
Visual field		
Pain with eye movement		
Fever		
Eye Pain/tenderness		

Preseptal Cellulitis

Description:

- Relatively common eyelid infection
- Involves the periorbital tissue _____ to the orbital septum
  - Acute eye lid:
    - Edema
    - Erythema

Orbital septum





Preseptal Cellulitis

- More common and less serious than orbital cellulitis
  - Generally mild; rarely leads to serious complications
    - *Orbital cellulitis may cause vision loss and even death!*
- Globe not affected
- Sometimes called periorbital cellulitis

Preseptal Cellulitis

Causes: 3 main sources

1. Direct inoculation through _____ such as:
   a. Laceration
   b. Strabismus surgery
   c. Insect bite
      - Typically causes infection by *Streptococcus aureus* or *Streptococcus pyogenes*







Preseptal Cellulitis



Causes: 3 main sources

2. Bacteria spread from nearby local _____ such as:
   a. Upper respiratory infection (URI)
   b. Sinusitis
   c. Dacryocystitis
   d. Acute hordeolum



Hordeolae



Dacryocystitis

SINUSITIS SYMPTOMS

ITCHY NOSE

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INFECTION

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SINUS

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CHILLS

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RHINITIS

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Preseptal Cellulitis

Causes: 3 main sources

3. Bacteria spread <u>hematogenously</u> from distant infection source such as:
   a. Otitis media
   b. Pneumonia



Preseptal Cellulitis

In general:

- Upper respiratory infection (URI) usually cause, if no evidence of trauma or local lid infection found
  - Recent history of URI with or without sinusitis is usually found



Preseptal Cellulitis

Risk factors:

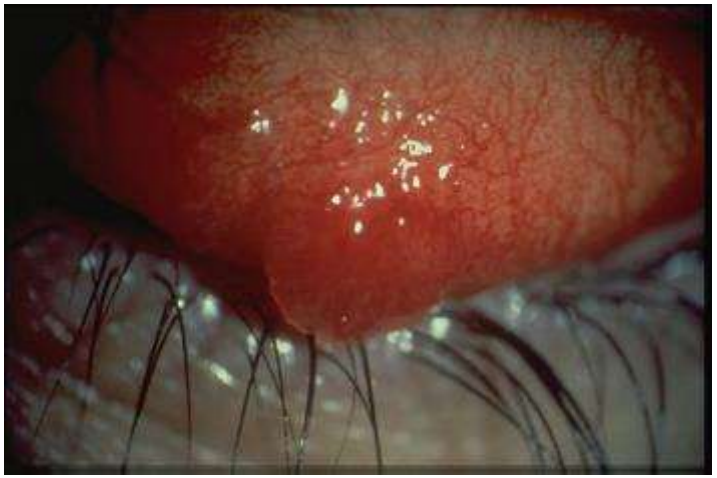
- Hordeola
- Chalazia
- Bug bites
- Trauma-related lesions
- Lesions caused by a recent surgical procedure near the eyelids
- Lesions caused by oral procedures
- Dacryocystitis

Hordeolum

- Aka stye:
  - Swollen eyelid
  - Sensitive to palpitation
- External hordeolum
  - Gland of Zeis infection
- Internal hordeolum
  - Meibomian gland infection
- *Staphylococcus aureus* is cause in 90-95% of cases
- Abscess = infectious pus



External Hordeolum



Internal Hordeolum

Chalazion

- Aka meibomian gland lipogranuloma or a meibomian cyst
- Painless
- Hard on palpitation
- No tenderness or inflammation



Preseptal Cellulitis

Offending Pathogens

Most common:

- *Staphylococcus aureus*
- *Staphylococcus epidermidis*
- *Streptococcus species*
- *Haemophilus influenzae*

Preseptal Cellulitis

- In the United States, the most common bacterial agent identified in *preseptal cellulitis* in **children** is now ***Streptococcus pneumoniae***

Israele V, Nelson JD: Periorbital and orbital cellulitis. Pediatr Infect Dis J 6:404, 1987

Adams WG, Deaver KA, Cochi et al: Decline of childhood Haemophilus influenzae type b (Hib) disease in the Hib vaccine era. JAMA 269:221, 1993

Preseptal Cellulitis

Offending pathogens

- Biologic warfare concern
  - Small pox
  - Anthrax

Preseptal Cellulitis

Other potential offenders

PC associate with or secondary to:

- Tuberculosis (TB) Ophthalm 2004;111 Raina et al
- Chlamydial conjunctivitis Head & Face medicine, 2007,
- Juvenile idiopathic arthritis (JIA)Eye 2003, 17, 258
- Acinetobacter Optometry 2005, 76(3) Miller J.
- Methicillin resistant Staphylococcus aureus (MRSA) bjo 2007

Preseptal Cellulitis

- You MUST Rule out Orbital Cellulitis!!!



Preseptal Cellulitis



Subjective findings:

- Red swollen eyelid
- Pain: Lid pain
  - If present, tends to be mild
- Epiphora
- Red Eye (Conjunctivitis)
- May have mild to moderate fever

Preseptal Cellulitis



Objective findings:

- Acute eyelid erythema
- Conjunctival injection
- Tenderness to touch
- Eyelid edema
- NO PUPIL DEFECT
- NO PROPTOSIS
- NO EOM RESTRICTION (ophthalmoplegia)
- NO PAIN ON EOMs
- NO DIPLOPIA
- Usually Normal Visual Acuities
- NO ONH SWELLING

Preseptal Cellulitis

Diagnosis:

- Careful examination of eyelid tissue for:
  - Puncture wound
  - Trauma
  - Eye lid infection
  - Infectious skin lesions



Preseptal Cellulitis

Diagnosis:

- Primarily pediatric disease
  - Approximately 80% patients younger than 10
    - In some studies



Preseptal Cellulitis

- Lid swelling and inflammation is present in BOTH preseptal and orbital disease.
  - Can be so severe, the patients cannot open the eye voluntarily.



Orbital cellulitis



Preseptal cellulitis

Preseptal Cellulitis

- Orbital CT necessary if eyelids **cannot be** separated to evaluate for:
  - Proptosis
  - Limited ocular motility
  - VA loss
  - ONH swelling



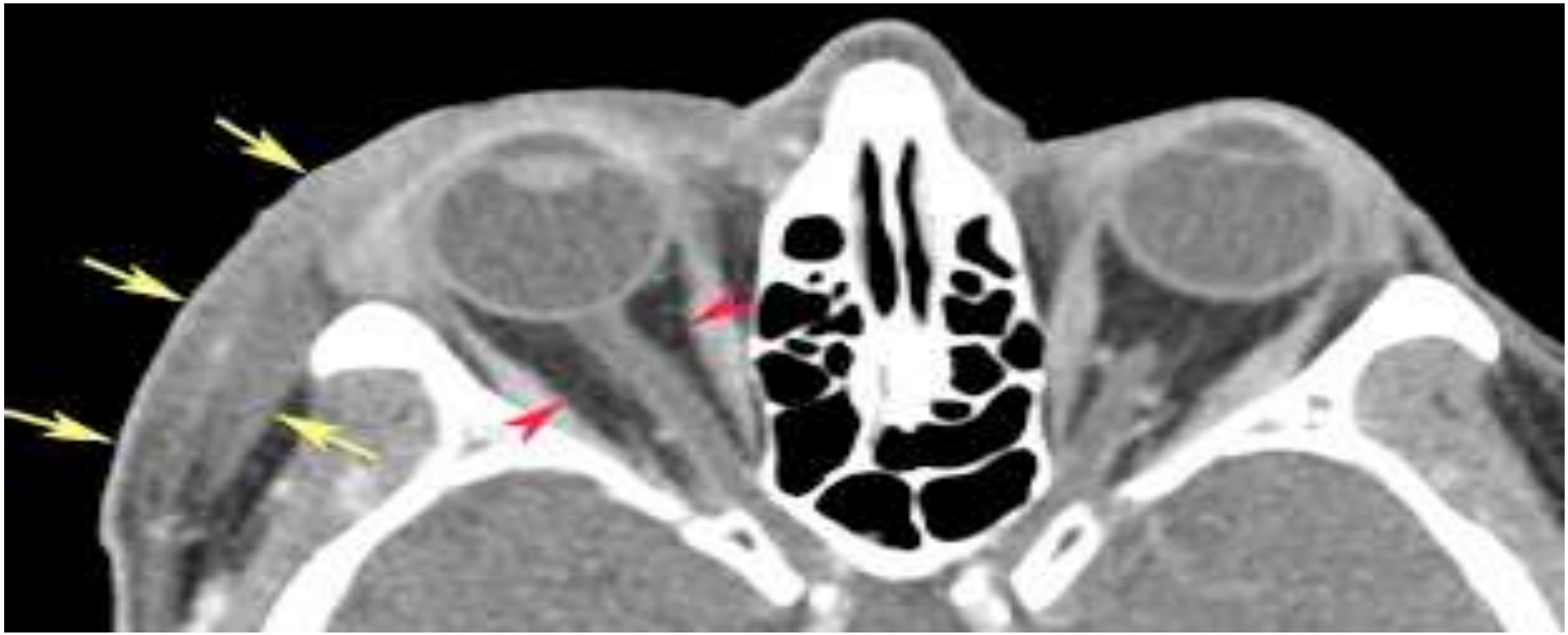
Preseptal Cellulitis

- CT scan recommended if:
  - Orbital involvement cannot be ruled out on clinical examination alone
  - Fever and leukocytosis
  - Disease progression despite antibacterial therapy
  - Young children where reliable exam is difficult



CT scan of preseptal cellulitis

Preseptal Cellulitis



- Soft tissue **thickening** of right preseptal region (between arrows).
- Retro-orbital fat is **normal** (red arrowheads). No orbital inflammation

Preseptal Cellulitis

Treatment: Empiric Tx for adults and older children

Amoxicillin/clavulanate (Augmentin)

- Mild to moderate: 500/125 mg PO TID
  - For 10 days
- Severe: 875/125 mg PO BID
  - For 10-14 days

- Must follow-up _____!!

Preseptal Cellulitis

Treatment: Mild preseptal cellulitis

- Amoxicillin/clavulanate (Augmentin)
  - Children older than 5 (<40Kg)
    - 25 to 45 mg/kg/day PO in two divided doses,
      - Maximum daily dose 90 mg/kg/day
  - Infants <3 months
    - 30 mg/kg PO divided q12h X 10 days
      - Consult with pediatrician on infants and children

Preseptal Cellulitis

- **Treatment Alternatives**
- Clindamycin:
  - Covers *Staph. aureus* (including MRSA), *S pneumoniae*, most other streptococci, and anaerobes but has poor *Haem. influenzae* coverage
  - **Adult:** 600 mg PO q8h for 10-14 days
  - **Pediatric**: 30-40 mg/kg/day PO divided q8h for 10-14 days (maximum 1.8 g/day)

Preseptal Cellulitis

- **Treatment (other option)**
- Ceftriaxone (Rocephin)
  - 3rd gen cephalosporin

Preseptal Cellulitis

Treatment: If allergic to PCN or exposure to MRSA

- Trimethoprim + sulfamethoxazole (Bactrim)
- Bactrim DS (trimethoprim 160mg + sulfamethoxazole 800 mg)
  - Adults
    - 1-2 tablets Q 12 hours X 10 days
  - Children (older than 5)
    - 8-12 mg/kg/day trimethoprim with 40-60 mg/kg/day sulfamethoxazole PO In two divided doses X10 days

Or

- Doxycycline 100 mg PO BID (contraindicated in children, pregnant women, nursing mothers)

Preseptal Cellulitis

Treatment:

Hospitalize for IV antibiotics if:

- Moderate to severe cellulitis
- Patient is toxic
- Patient noncompliant
- Child younger than 5 years
- No noticeable improvement or _____ after a few days with Tx



Preseptal Cellulitis

Treatment:

- Examples of IV antibiotics:
  - Ampicillin + sulbactam (Unasyn)
  - Ceftriaxone (e.g. Rocephin)
  - Vancomycin, if methicillin-resistant S. aureus (MRSA) is suspected.

Preseptal Cellulitis

- **Treatment:**
- Warm compresses to inflamed area TID-PRN
- Polymyxin B/bacitracin ointment to eye QID if secondary conjunctivitis present
- Nasal decongestants if sinusitis present

Case

- 10-year-old presenting with fever, acute unilateral eyelid erythema, and limited extraocular motions
- Presentation suspicious for orbital cellulitis

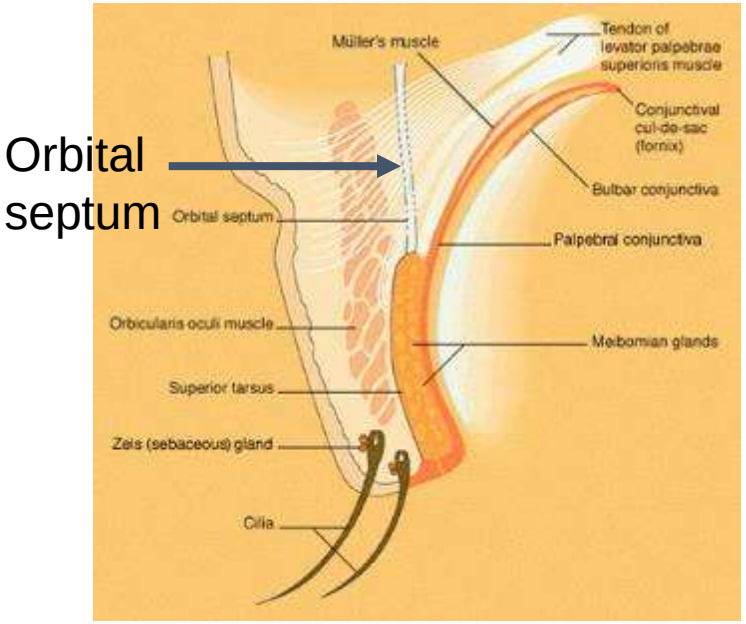


Orbital Cellulitis



Description:

- Infection of orbital soft tissue _____ to the orbital septum
  - Involves fat and muscles
- Potentially vision and _____
- Less common than preseptal cellulitis



Orbital Cellulitis

Description:

- More common in children than adults
  - Median age of children hospitalized with orbital cellulitis is 7-12 years
  - In children
    - 2:1 males: females

Orbital Cellulitis

Description:

- In adults,
  - Equal in males and females
    - Except for cases caused by MRSA
      - More common in females by 4:1 ratio.
- Tends to occur in winter due to:
  - Increased incidence of sinusitis in cold weather

Orbital Cellulitis



Causes:

1. Uncommon bacterial sinusitis complication
   a. Cause of most cases of orbital cellulitis
2. Extension of infection from periorbital structures, sinuses and teeth
3. Direct infection of orbit following:
   1. Trauma or surgery of eyelid, strabismus or retinal surgery
4. Hematogenous spread of bacteremia

Orbital Cellulitis

- **Most** cases caused by ethmoid sinusitis
- Must ask about **recent _____ work**
- 11% of cases result in vision loss
- If not treated can cause sinus thrombosis (1%) and meningitis (2%) which can be life threatening!!!



Orbital Cellulitis

Offending Pathogens

Most common bacterial ones:

- *Streptococcus species*
- *Staphylococcus aureus*
- *Haemophilus influenzae* type B
- *Strep. pyogenes*

Orbital Cellulitis

Offending Pathogens:

Most common fungal:

- *Mucor*
- *Aspergillus*

Orbital Cellulitis

Subjective:

- Red, swollen eye/eyelid
- Fever
- Malaise
- History of recent URI or sinusitis
- Headache, sinus pain
- Tooth pain
- Recent history of facial trauma, surgery, dental work



Orbital Cellulitis

Subjective:

- Red eye
- Decreased vision
- Pain on eye movement
- EOM Restriction***, ophthalmoplegia
- Proptosis***



Orbital Cellulitis



Chemosis, proptosis, ophthalmoplegia, and edema and erythema of eyelids

Objective:

- Double vision
- Orbital pain, tenderness
  - Presents early
- Lid edema, erythema
- Conjunctival chemosis, hyperemia
- May have elevated IOP

Orbital Cellulitis



Objective:

- Red discoloration of eyelid
- Nasal discharge may be present
- Optic disc edema
- Afferent Pupillary Defect (+ APD)
- Ophthalmoplegia and Proptosis are key findings of orbital cellulitis
  - Proptosis may be subtle

Orbital Cellulitis

- **Work up:**
- Case History!!!
  - Trauma or surgery?
  - Ear, nose, throat, or systemic infection?
  - Tooth pain or recent dental abscess?
  - Stiff neck or mental status changes?
  - Diabetes or immunosuppressive illness?

Orbital Cellulitis

Work up:

- Complete ophthalmic examination: Look for
  - Afferent pupillary defect (APD)
  - Limitation of or pain with EOMs
  - Proptosis
  - Elevated IOP
  - Decreased color vision
  - Optic nerve or fundus abnormality

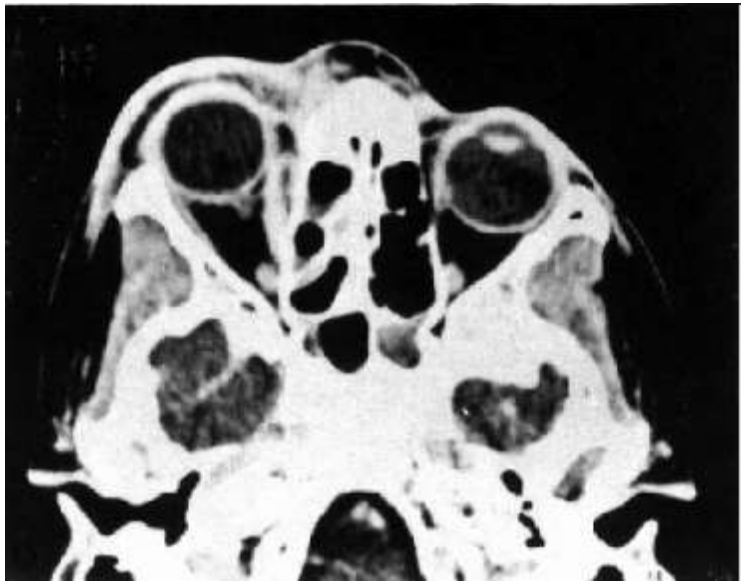
- Check vital signs, mental status, and neck flexibility

Orbital Cellulitis

Work up:

- Lab workup
  - CBC: Count > 15,000
  - Blood culture
  - Culture material from sinuses or orbital abscess
  - Needle aspiration of orbit <u>contraindicated</u>
- Lumbar puncture advisable if cerebral or meningeal signs develop

Orbital Cellulitis



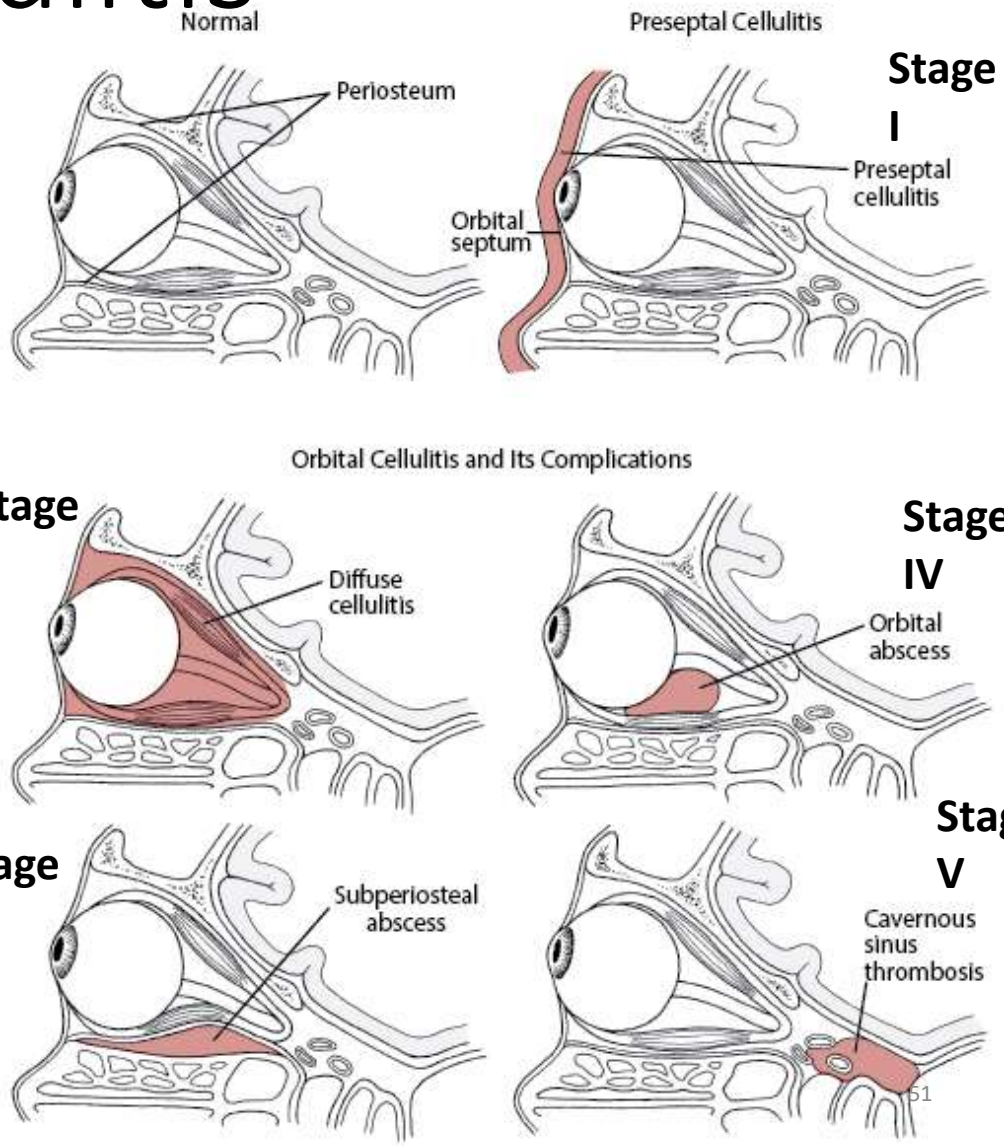
Work up: Imaging

- Imaging ESSENTIAL!!!
  - Hi resolution CT scan with contrast with axial and coronal views
    - Clinicians may be reluctant to obtain CT scan on pediatric patients due to concern of radiation exposure
  - MRI may be needed to evaluate for cavernous sinus disease

Orbital Cellulitis

Orbital involvement on CT

- Stage I - Preseptal cellulitis
- Stage II - Inflammatory orbital edema
- Stage III - Subperiosteal abscess
- Stage IV - Orbital abscess
- Stage V - Cavernous sinus thrombosis



Orbital Cellulitis

Treatment:

- Hospitalization!!! All patients should be promptly hospitalized until improvement clear
  - Medical management
    - IV broad spectrum antibiotics are started STAT until pathogen identified
      - For 1-2 weeks followed by oral Tx for 2-3 weeks
    - Ampicillin-sulbactam is good initial choice for broad-spectrum coverage

Orbital Cellulitis

Treatment:

- Hospitalization!!!
  - Medical management:
    - Penicillin G, cefoxitin, metronidazole, clindamycin, chloramphenicol for suspected anaerobic infections
    - If MRSA suspected then
      - Vancomycin, cefotaxime, trimethoprim + sulfamethoxazole, clindamycin

Orbital Cellulitis

- Fungal orbital cellulitis potentially lethal
  - Principal organisms:
  - *Mucor* and *Aspergillus*
  - Tx with antifungals
    - Amphotericin B deoxycholate (AmBisome)

Orbital Cellulitis

Treatment: Hospitalization!!!

- **Surgical drainage IF**:
  - Poor response to appropriate Tx in 2-3 days
  - Decrease in VAs
  - Positive APD develops
  - Progression of proptosis despite treatment
  - Abscess size doesn't decrease in 2-3 days
  - Brain abscess develops and doesn't respond to AB Tx. Then craniotomy indicated

Chai-Lee et al, Image-guided system endoscopic drainage of orbital abscess caused by methicillin-resistant Staphylococcus aureus in an infant, International Journal of Surgery Case Reports Volume 33, 2017, Pages 119-123



Orbital Cellulitis

- Consultation
  - Pediatrician, internist, family physician
  - Infectious disease specialist
  - Ear-nose-throat (ENT) specialist if sinusitis is causative factor

Summary

- Preseptal cellulitis is more common than Orbital cellulitis
- Proptosis and ophthalmoplegia can aid in DDx the two
- Most preseptal cellulitides will respond to PO antibiotics on an outpatient basis
- Orbital cellulitis needs hospitalization, IV antibiotics, imaging
- Antibiotics should cover Gram+ and Gram-

Comparison Table

Clinical presentation	PRESEPTAL	ORBITAL
Eyelid edema	Present	Almost always present
Chemosis (swelling of conj)	Typically absent	May be present
Pupils/APD	Normal	May be affected
Extraocular movements (EOM) +/- diplopia	Intact	May be restricted; patient may have pain & diplopia on EOMs
Visual acuity	Rarely affected	May be markedly decreased
Proptosis	Absent	May be present
Color vision	Intact	May be diminished
Visual fields	Intact	May be constricted
Pain with eye movement	No	Present
Fever	Maybe present	Usually present
Eye Pain/tenderness	Maybe present	Yes, maybe associated with deep pain

Autoimmune Diseases Of Lids and Conjunctiva

Poliosis

- Description:
  - Localized patch of white hair
  - Can affect any hairy area
  - Decreased or absent melanin in hair bulb or follicle





Poliosis

Poliosis of the eyelashes & eyebrows



Etiology:

- Can occur with various genetic or acquired conditions
- **Genetic syndromes**:
  - Piebaldism
    - Congenital absence of melanocytes of skin and hair,
      - Eyes rarely affected
  - Waardenburg
    - Rare genetic disorder characterized by sensorineural deafness and pigmentary abnormalities
  - Tuberous sclerosis (TSC)
    - Seizures and dermatologic manifestation at early age

Poliosis circumscripta: overview and underlying causes, Sleiman R et al J Am Acad Dermatol. 2013 Oct;69(4):625-33

Poliosis

Etiology:

- **Acquired conditions**:
  - Inflammatory conditions
  - Neoplastic entities - benign and malignant
    - Mainly melanocytic
  - Medications
  - Others



Poliosis circumscripta: overview and underlying causes, Sleiman R et al J Am Acad Dermatol. 2013 Oct;69(4):625-33

Poliosis



Poliosis of eyelashes and eyebrows

Etiology:

- **Acquired conditions (continued):**
  - **Vitiligo**
    - Chronic polygenic pigmentary disorder leading to depigmented macules
  - **Vogt-Koyanagi-Harada (VKH) syndrome**
    - T-cell–mediated autoimmune inflammatory disorder
  - **Alopecia:**
    - hair loss
  - **Sarcoidosis:**
    - Inflammatory Dz affecting multiple body organs
  - **Blepharitis**

Poliosis circumscripta: overview and underlying causes, Sleiman R et al J Am Acad Dermatol. 2013 Oct;69(4):625-33

Poliosis

Etiology:

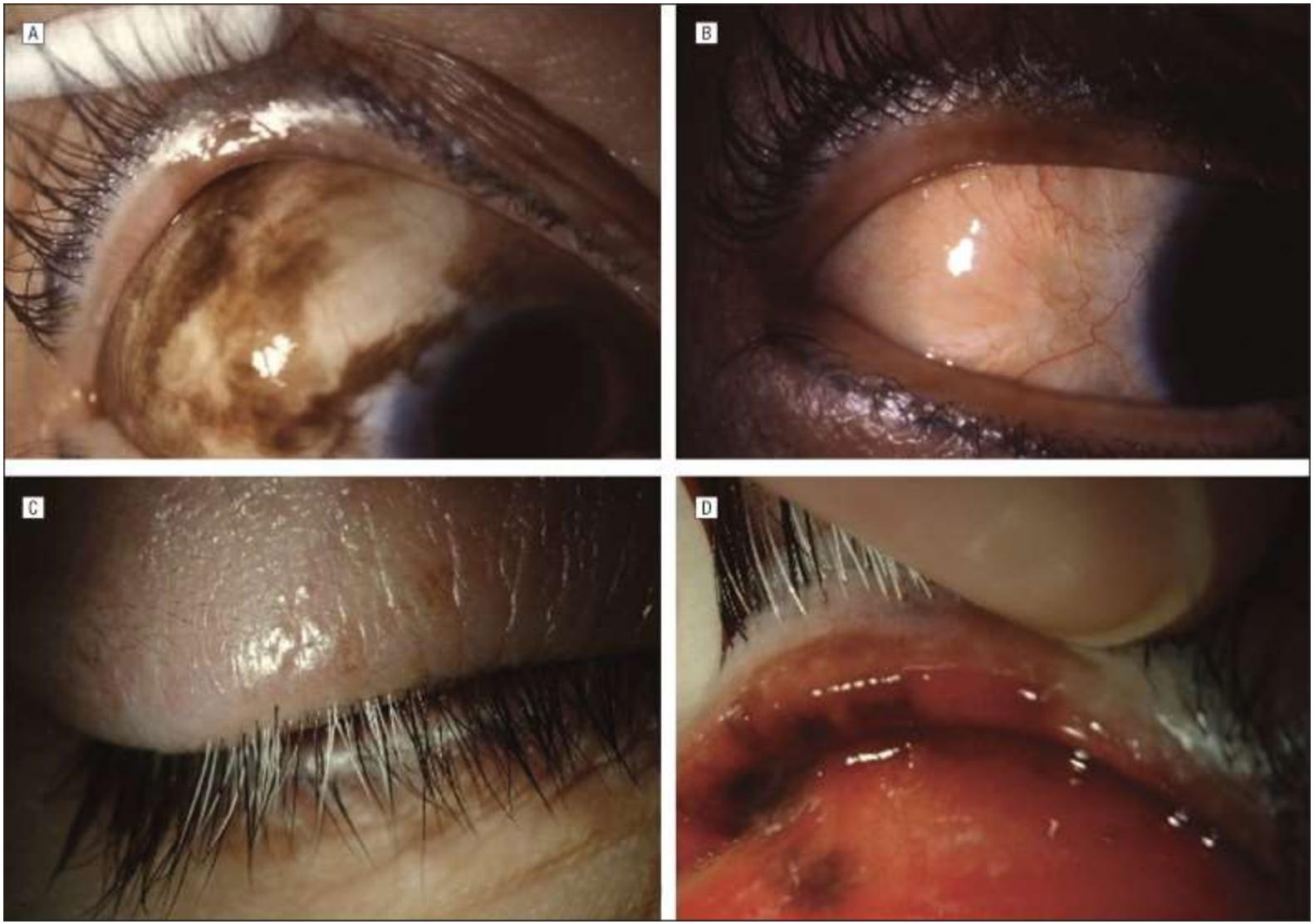
- Poliosis associated with neoplastic lesions
  - **Melanocytic lesions**
    - Both benign and malignant
    - May be first visible sign of melanoma



A 60-year-old man had a nontender, firm mass in the left anterior orbit superonasally and poliosis of the left upper eyelid lashes for 3 months

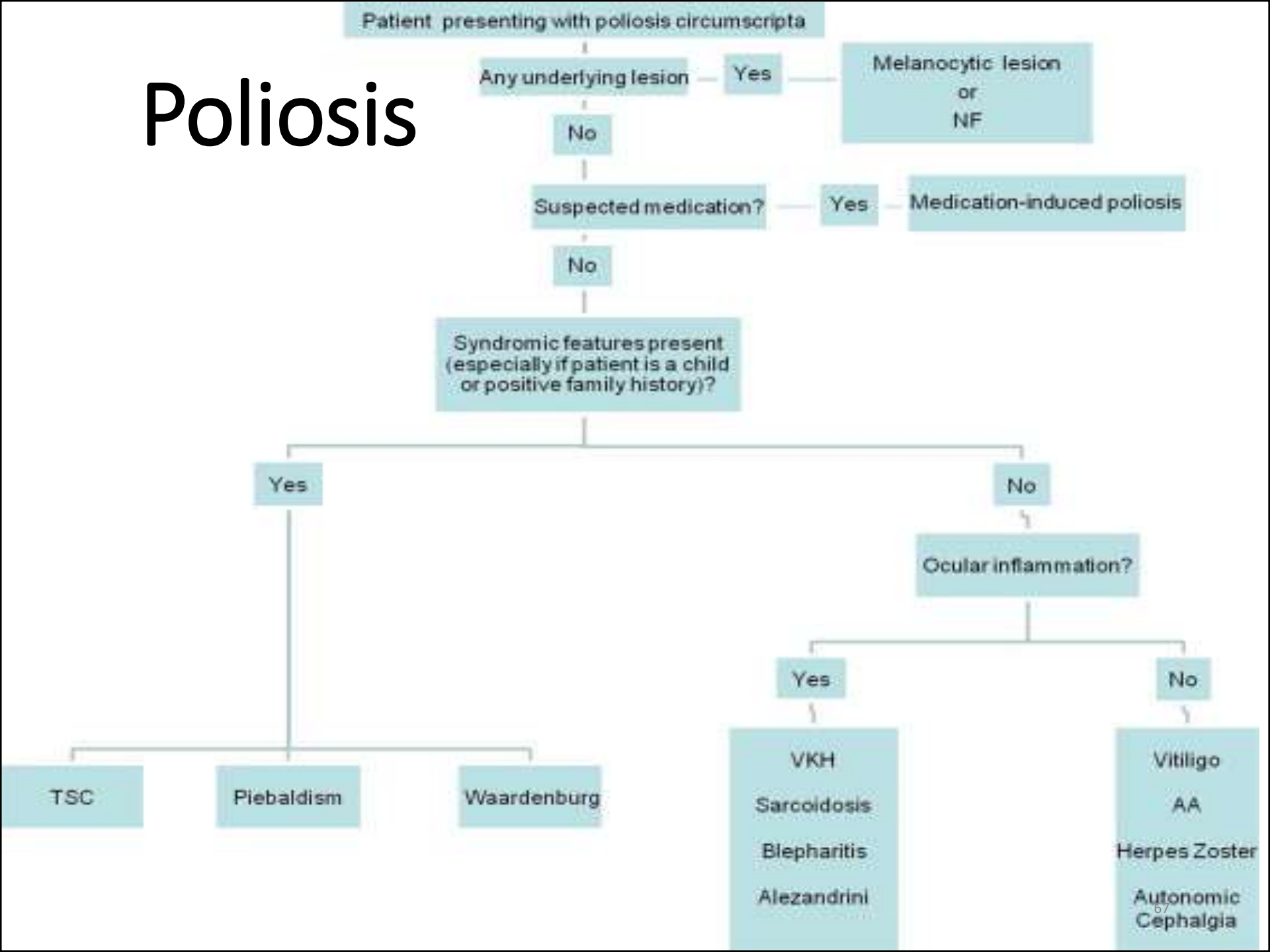
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Poliosis



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Poliosis


Patient presenting with poliosis circumscripta
Any underlying lesion
Yes
Melanocytic lesion
or
NF
No
Suspected medication?
Yes
Medication-induced poliosis
No
Syndromic features present (especially if patient is a child or positive family history)?
Yes
No
Ocular inflammation?
Yes
No
TSC
Piebaldism
Waardenburg
VKH
Sarcoidosis
Blepharitis
Alezandrini
Vitiligo
AA
Herpes Zoster
Autonomic Cephalgia

Poliosis



- Plan/TX
  - Treat blepharitis or other underlying cause
  - Consultation and referral if melanoma or systemic condition suspected



Blepharitis

Vitiligo

Autoimmune disorder

- Acquired pigmentary disorder of skin
- Usually bilateral eye lid presentation
- Thyroid abnormalities most common
- Frequently positive family history
  - 30% of cases



Vitiligo on face and lids

Vitiligo

- Description:
  - Usually presents before age 20
  - Progressive
  - Asymptomatic
  - No racial predilection
    - More obvious in darker skin



Vitiligo



- A case of Vitiligo????
  - 1-2% of the world's population have it
  - Facial lesions may favor
    - Periocular or
    - Perioral distribution



Vitiligo

Lesion characteristics:

- White or depigmented macules and patches
- Round, oval, or linear in shape
- Well demarcated
- Range from millimeters to centimeters in size
- Enlarge outward over time
- Unpredictable rate of growth



Vitiligo



- Plan/TX
  - Determine if family history
  - R/O associated syndromes
  - Cosmetic camouflage
  - Skin protection from sun!!
  - Melanocyte stimulation and skin color blending with amino acids

Vitiligo

- Plan/TX
  - Topical steroids
  - Dermatological consult
    - PUVA - UV light Tx
      - Combines a plant-derived substance called psoralen with light therapy (photochemotherapy)
    - Narrow band UVB Tx:
      - Been shown to stop or slow the progression of active vitiligo.
    - Grafting
    - Depigmentation
  - Others

THANK YOU!